

Q1

$$y = 3 \cos^2 \left(x + \frac{\pi}{6} \right) - 2$$
$$\frac{dy}{dx} = -6 \cos \left(x + \frac{\pi}{6} \right) \sin \left(x + \frac{\pi}{6} \right)$$

$$x + \frac{\pi}{6} = \arccos 0$$
$$x + \frac{\pi}{6} = \frac{\pi}{2}$$
$$x = \frac{\pi}{2} - \frac{\pi}{6}$$
$$x = \frac{\pi}{3}$$